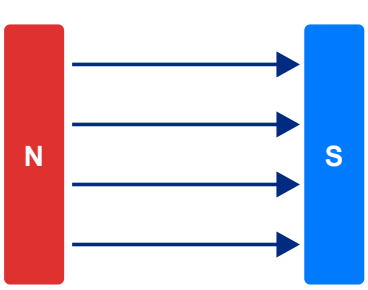
A2 · Unit 4 · WPH14 Further Mechanics, Fields & Particles

Magnetic Fields & Induction

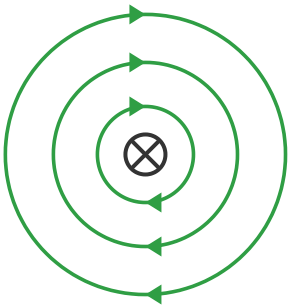
A magnetic field pushes sideways on moving charges, and that one fact explains electric motors, the curved tracks in particle detectors and the beam steering in old TV tubes. Run it backwards and you get electromagnetic induction: move a magnet near a coil and a voltage appears from nowhere. That is how almost all of the world's electricity is generated. Let's build both ideas up from the force on a single wire.

Magnetic fields and flux density

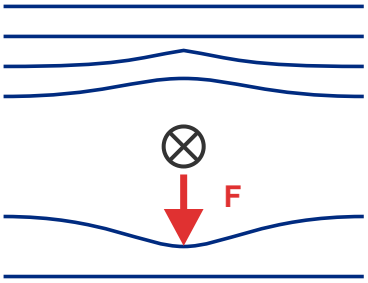
A **magnetic field** is a region where a magnet, a current-carrying wire or a moving charge feels a force. We draw it with field lines that run from north to south outside a magnet. The closer together the lines, the stronger the field. Between two flat poles the lines are straight, parallel and evenly spaced: a **uniform** field. A current in a straight wire makes circular field lines around it, in the direction given by the **right-hand grip rule** (thumb along the current, fingers curl the way the field goes).



uniform field between poles
(B the same everywhere)



wire, current into the page
(right-hand grip: clockwise)



lines bunch above, thin out below:
the wire is pushed down

Above the wire its own field points the same way as the magnet's field and they add; below, they point opposite ways and partly cancel. The wire is pushed from the strong-field side towards the weak-field side.

The strength of a magnetic field is measured by its **magnetic flux density** B. It is defined by the force it exerts on a wire carrying a current at right angles to the field: the bigger the force on each metre of wire for each amp of current, the stronger the field.

DEFINITION

Magnetic flux density B is the force per unit length per unit current on a conductor placed at right angles to the field, $B = F/Il$. Its unit is the **tesla**: 1 T is the flux density that exerts a force of 1 N on a 1 m length of

wire carrying 1 A at right angles to the field ($1\ \text{T} = 1\ \text{N}\ \text{A}^{-1}\,\text{m}^{-1}$). B is a vector, in the direction of the field lines.

For scale: the Earth's field is about 5×10^{-5} T, a fridge magnet about 0.01 T, a strong laboratory magnet about 1 T and an MRI scanner 1.5 to 3 T.

Force on a current-carrying wire: F = BIl sin θ

The force on a wire in a magnetic field is proportional to the flux density, the current and the length of wire in the field. It also depends on the angle θ between the wire and the field. Only the part of the current that cuts **across** the field lines feels a force, so a wire lying along the field feels nothing at all.

Force on a current-carrying conductor

FORMULA SHEET

$$F = BIl\sin\theta$$

F: force / N B: flux density / T I: current / A l: length in the field / m

θ: angle between wire and field

The force is always at right angles to **both** the field and the current. To find which way, use **Fleming's left-hand rule**: hold your thumb, first finger and second finger of your left hand at right angles to one another.

- **F**irst finger: **F**ield (north to south)
- se**C**ond finger: **C**urrent (conventional, + to −)
- thu**M**b: **M**otion, the direction of the force

EXAM TIP

In Fleming's rule the current finger must be at right angles to the field finger. If the wire is at an angle, use the component of the current at right angles to the field. That is exactly what the $\sin\theta$ in the equation does.

WORKED EXAMPLE

A straight wire carries a current of 2.5 A. A 6.0 cm length of it lies in a uniform magnetic field of flux density 0.15 T, at an angle of 40° to the field. Calculate the force on the wire. What is the largest force this wire could feel in this field, and how would you arrange it?

Show solution

$F = BIl\sin\theta = 0.15 \times 2.5 \times 0.060 \times \sin 40^\circ = 0.0225 \times 0.643 = 1.4 \times 10^{-2}$ N

The largest force is when $\sin\theta = 1$, with the wire at 90° to the field: $F = 0.15 \times 2.5 \times 0.060 = 2.3 \times 10^{-2}$ N. The direction is perpendicular to both the wire and the field, from Fleming's left-hand rule.

Force on a moving charge: F = Bqv sin θ

A current is just moving charges, so the force on a wire must really be the sum of forces on all the charge carriers inside it. A single charge moving through a field feels a force too, even with no wire at all.

From F = BIl to F = Bqv

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Take a wire of cross-sectional area A with n charge carriers per unit volume, each of charge q, drifting at speed v. From Unit 2, the current is:

$$I = nqvA$$

2. With the wire at right angles to the field, the force on a length l is:

$$F = BIl = B(nqvA)\,l$$

3. The number of charge carriers in that length of wire is (number per unit volume) × (volume):

$$N = nAl$$

4. Divide the total force by the number of carriers to get the force on each one:

$$\frac{F}{N} = \frac{B\,nqvA\,l}{nAl} \quad \Rightarrow \quad \boxed{F = Bqv}$$

If the charge moves at an angle θ to the field, only the velocity component $v\sin\theta$ across the field counts, giving $F = Bqv\sin\theta$.

Force on a moving charge

FORMULA SHEET

$$F = Bqv\sin\theta$$

q: charge / C v: speed / m s^{-1} θ: angle between velocity and field

Fleming's left-hand rule still works, but the second finger points along the **conventional current**. For a positive charge that is the way it moves. For an electron (or any negative charge) the current points **opposite** to its velocity, so the force is reversed.

Circular motion in a uniform field

The magnetic force is always at right angles to the velocity. That has two big consequences:

- It does **no work** on the particle (force × distance moved in the direction of the force is zero), so the particle's speed and kinetic energy never change.
- It changes the particle's direction at a steady rate, so in a uniform field at right angles to its motion the particle moves in a **circle**. The magnetic force provides the centripetal force.

Setting the magnetic force equal to the centripetal force, $Bqv = mv^2/r$, gives the radius of the circle, $r = mv/BQ = p/BQ$. That result is **derived** in full on the Nuclear & Particle Physics page, where it is used to measure the momentum of particles from their curved tracks.

WORKED EXAMPLE

An electron moves at $3.0 \times 10^7\ \text{m s}^{-1}$ at right angles to a uniform magnetic field of flux density 4.0 mT. Calculate the magnetic force on it and its acceleration.

Show solution

$F = Bqv = 4.0 \times 10^{-3} \times 1.60 \times 10^{-19} \times 3.0 \times 10^7 = 1.9 \times 10^{-14}\ \text{N}$

$a = F/m = 1.92 \times 10^{-14}/9.11 \times 10^{-31} = 2.1 \times 10^{16}\ \text{m s}^{-2}$, directed towards the centre of its circular path. Tiny force, enormous acceleration, because the electron's mass is so small.

Magnetic flux and flux linkage

To talk about induction we need a way to say "how much field passes through a coil". Picture the field lines threading through a loop of wire. If the loop faces the field, many lines pass through; if it is tilted, fewer do; if it lies edge-on to the field, none do. **Magnetic flux** Φ measures this. For an area A at right angles to a uniform field, $\Phi = BA$. That is also why B is called flux **density**: it is flux per unit area.

Magnetic flux through an area

$$\Phi = BA\cos\theta$$

Φ: flux / Wb A: area / m² θ: angle between the field and the normal to the area

The $\cos\theta$ is there because only the component of B at right angles to the area threads through it. When the area faces the field squarely, $\theta = 0$ and $\Phi = BA$.

A coil of N turns has the same flux through every turn, and each turn will contribute to the induced e.m.f., so we multiply: the **flux linkage** is $N\Phi$.

DEFINITION

Magnetic flux Φ through an area A at right angles to a uniform field B is $\Phi = BA$. Its unit is the **weber**, $1\ \text{Wb} = 1\ \text{T m}^2$.

Flux linkage is the flux through a coil multiplied by the number of turns, $N\Phi$. Its unit is also the weber (sometimes written Wb-turns).

Electromagnetic induction: a magnet and a coil

Push a magnet into a coil connected to a sensitive meter and the needle kicks. Hold the magnet still inside the coil and the needle returns to zero. Pull it out and the needle kicks the other way. An e.m.f. is **induced** in the coil, but only while the flux linkage through it is **changing**.

Only the **relative** motion matters. Seen from the magnet, it is the wire that moves through the field, so its free electrons feel the Bqv force and are pushed along the wire: the wire is "cutting" field lines. Stop the motion and the push stops. Whether you move the magnet or the coil, the e.m.f. is the same.

KEY IDEA

Factors affecting the e.m.f. induced by a moving magnet

The size of the induced e.m.f. is **bigger** if you:

- move the magnet (or the coil) **faster**, so the flux linkage changes more quickly;
- use a **stronger magnet**, so the change in flux is bigger;
- use a coil with **more turns**, so the flux links more turns;
- use a coil with a **larger cross-sectional area** (more flux through each turn).

The **direction** of the e.m.f. reverses if you reverse the motion (in instead of out) or turn the magnet round (south pole first). There is no e.m.f. at all when there is no relative motion.

COMMON MISTAKE

The induced **e.m.f.** does not depend on the resistance of the circuit, but the induced **current** does ($I = \varepsilon/R$). If the coil is not part of a complete circuit there is still an e.m.f. across its ends, but no current flows.

Faraday's law and Lenz's law

The simulation above shows the pattern: the e.m.f. is large where the flux linkage graph is steep and zero where it is flat. The e.m.f. is the **rate of change of flux linkage**. That is **Faraday's law**. The direction comes from **Lenz's law**, and the two combine into one equation.

DEFINITION

Faraday's law: the magnitude of the induced e.m.f. is equal to the rate of change of flux linkage.

Lenz's law: the direction of the induced e.m.f. (and of any induced current) is such as to oppose the change that produces it.

Faraday's and Lenz's laws

FORMULA SHEET

$$\varepsilon = -\frac{d(N\Phi)}{dt}$$

ε: induced e.m.f. / V $N\Phi$: flux linkage / Wb t: time / s

$d(N\Phi)/dt$ is the gradient of a flux linkage–time graph. If the flux linkage changes steadily, the e.m.f. is just $\Delta(N\Phi)/\Delta t$. If it does not, that ratio gives the **average** e.m.f., and the instantaneous e.m.f. is the gradient of the tangent. The minus sign is Lenz's law: the e.m.f. acts to **oppose** the change in flux linkage. In calculations of the size of the e.m.f. you can leave it out.

KEY IDEA

Why Lenz's law has to be true: energy conservation

Suppose the induced current helped the change instead of opposing it. Push a magnet towards a coil; the induced current would attract the magnet further in, so it would speed up, which would induce a bigger

current, which would pull harder... The magnet would accelerate and the coil would heat up, all from one small push. That would create energy from nothing.

So the induced current must **oppose** the motion. You have to do work pushing the magnet against the repulsion, and that work is exactly the electrical energy that appears in the circuit. Lenz's law is the conservation of energy applied to induction.

A rod moving through a field

The cleanest case of all is a straight conductor moving at right angles to a uniform field, like an aircraft wing or a metal rod sliding along rails.

The e.m.f. across a moving rod, ε = Blv

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. A rod of length l moves at speed v at right angles to a uniform field B. In a time Δt it moves a distance $v\,\Delta t$, sweeping out an area:

$$\Delta A = l\,v\,\Delta t$$

2. The flux it cuts in that time (one "turn", so $N = 1$) is:

$$\Delta\Phi = B\,\Delta A = Blv\,\Delta t$$

3. Faraday's law: the e.m.f. is the rate of change of flux linkage:

$$\varepsilon = \frac{\Delta\Phi}{\Delta t} = \frac{Blv\,\Delta t}{\Delta t} \quad \Rightarrow \quad \boxed{\varepsilon = Blv}$$

4. Check it with forces instead. Each free electron in the rod moves with it, so feels $F = Bev$ along the rod. Electrons pile up at one end until the electric field this creates balances the magnetic push, $eE = Bev$. With $E = V/l$:

$$e\frac{V}{l} = Bev \quad \Rightarrow \quad V = Blv$$

Same answer: "cutting flux" and "the force on moving charges" are two descriptions of the same physics.

WORKED EXAMPLE

An aircraft with a wingspan of 35 m flies horizontally at 240 m s^{-1} where the vertical component of the Earth's magnetic field is 4.5×10^{-5} T. Calculate the e.m.f. induced between its wingtips. Explain why this e.m.f. could not be used to light a lamp in the cabin.

Show solution

The wings cut the vertical component of the field: $\varepsilon = Blv = 4.5 \times 10^{-5} \times 35 \times 240 = 0.38$ V

Any wires connecting the wingtips to a lamp travel with the aircraft through the same field, so the same e.m.f. is induced in them, in the same direction round the loop. The two e.m.f.s cancel, the flux linkage of the complete circuit does not change, and no current flows.

A coil rotating in a field

Rotate a flat coil in a uniform field and the angle between the field and the coil keeps changing, so the flux linkage rises and falls. Here you can see Faraday's law as a graph: the e.m.f. at each instant is minus the gradient of the flux linkage graph.

COMMON MISTAKE

A very common error is to say the e.m.f. is biggest when the flux linkage is biggest. It is the other way round for a rotating coil: the e.m.f. depends on how fast the flux linkage is **changing**, not on how big it is.

WORKED EXAMPLE

A coil of 250 turns and area 3.0 cm² lies with its plane at right angles to a uniform field of 0.40 T. The field is switched off and falls steadily to zero in 20 ms. (a) Calculate the e.m.f. induced. (b) The coil is part of a circuit of total resistance 5.0 Ω. Calculate the current and the total charge that flows.

Show solution

(a) Initial flux linkage: $N\Phi = NBA = 250 \times 0.40 \times 3.0 \times 10^{-4} = 3.0 \times 10^{-2}$ Wb. It falls to zero, so $\Delta(N\Phi) = 3.0 \times 10^{-2}$ Wb.

$$\varepsilon = \frac{\Delta(N\Phi)}{\Delta t} = \frac{3.0 \times 10^{-2}}{20 \times 10^{-3}} = 1.5 \text{ V}$$

(b) $I = \varepsilon / R = 1.5 / 5.0 = 0.30$ A, flowing for 20 ms, so $Q = It = 0.30 \times 0.020 = 6.0 \times 10^{-3}$ C.

Notice that $Q = \Delta(N\Phi)/R$: switching the field off twice as fast would double the current but halve the time, so the same charge flows.

Mutual induction: two linked coils

You don't need a moving magnet to change the flux through a coil. Put two coils side by side. A current in the first (the **primary**) makes a magnetic field, and some of that field threads through the second (the **secondary**). Change the current in the primary and the flux linkage of the secondary changes, so an e.m.f. is induced in it, even though nothing moves and the coils are not electrically connected.

KEY IDEA

Factors affecting the e.m.f. induced by a changing current in a linked coil

- The **rate of change of the current** in the primary: switching on or off quickly, or changing the current quickly with a rheostat, gives a bigger e.m.f. A steady current, however large, gives none.
- The **size of the change** in current: a bigger current makes a bigger change in flux when it is switched.
- The **number of turns** on the secondary (and on the primary, which sets how much field each amp makes).
- How much of the primary's flux **links** the secondary: putting the coils closer and on the same axis, or winding both on a shared **soft iron core**, which concentrates the flux and carries it from one coil to the other.

By Lenz's law, while the primary current is **growing** the secondary's induced current makes a field that opposes the primary's field. While it is **falling**, the secondary's field points the same way as the primary's, trying to keep the flux up.

EXAM TIP

Exam questions often show a primary current–time graph and ask you to sketch the secondary e.m.f. Use $\varepsilon \propto -\text{gradient}$: wherever the current graph is flat (constant current, even if large), the e.m.f. is zero; where it is a straight sloping line, the e.m.f. is a constant; steeper slopes give bigger e.m.f.s; and the sign flips when the slope changes from up to down.

WORKED EXAMPLE

The flux linkage of a secondary coil rises steadily from 0 to 0.090 Wb in 5.0 ms when the primary is switched on, stays constant for 20 ms, then falls steadily back to zero in 2.0 ms when the primary is switched off. Calculate the e.m.f. in each stage and sketch the e.m.f.–time graph.

Show solution

Rising: $\varepsilon = -\dfrac{0.090}{5.0 \times 10^{-3}} = -18\ \text{V}$ for 5.0 ms.

Constant: the gradient is zero, so $\varepsilon = 0$ for 20 ms.

Falling: $\varepsilon = -\dfrac{-0.090}{2.0 \times 10^{-3}} = +45\ \text{V}$ for 2.0 ms.

The graph is a rectangle of −18 V lasting 5 ms, a gap at zero, then a taller, narrower rectangle of +45 V lasting 2 ms. Both rectangles have the same area (0.090 V s = 0.090 Wb), because the flux linkage returns to where it started.

Quick checks

QUICK CHECK

Which of these is equivalent to the tesla?

- N A m
- kg m s⁻² A⁻¹
- kg s⁻² A⁻¹
- Wb m²

From $B = F/Il$: $\text{T} = \text{N A}^{-1}\text{m}^{-1} = (\text{kg m s}^{-2})\,\text{A}^{-1}\text{m}^{-1} = \text{kg s}^{-2}\text{A}^{-1}$. Also $\text{T} = \text{Wb m}^{-2}$, not Wb m².

QUICK CHECK

An electron moves from left to right across the page through a uniform magnetic field directed into the page. Which way is the magnetic force on it?

- Towards the top of the page
- Towards the bottom of the page
- Into the page
- There is no force, because the electron moves at right angles to the field

The conventional current is opposite to the electron's motion, so it points right to left. First finger into the page, second finger to the left: the thumb points down the page. (A proton moving the same way would be pushed up.)

QUICK CHECK

A proton moves in a circle in a uniform magnetic field. How much work does the magnetic force do on it in one complete orbit?

- None, because the force is always perpendicular to the velocity
- Bqv × 2πr
- None, because the proton returns to its starting point, but some is done during each half orbit
- It depends on the speed of the proton

Work = force × displacement in the direction of the force. The force always points to the centre while the motion is along the circle, so no work is done at any instant. That is why the speed stays constant.

QUICK CHECK

A flat coil rotates steadily in a uniform magnetic field. At which position is the induced e.m.f. largest?

- When the plane of the coil is at right angles to the field, so the flux linkage is maximum
- When the plane of the coil is parallel to the field, so the flux linkage is zero
- When the coil is at 45° to the field
- The e.m.f. is the same at every position

The e.m.f. is the rate of change of flux linkage, the gradient of the $N\Phi$–t graph. That is steepest where $N\Phi$ passes through zero, and flat at the maximum.

QUICK CHECK

A bar magnet is dropped vertically through a coil connected to a data logger. Which describes the e.m.f.–time graph?

- Two identical pulses in the same direction
- Two pulses in opposite directions, the second smaller and wider
- Two pulses in opposite directions, the second taller and narrower
- A single pulse while the magnet is inside the coil

The flux linkage rises as the magnet enters and falls as it leaves, so the e.m.f.s are opposite. The magnet accelerates, so it leaves faster than it enters: the flux changes faster (taller pulse) for a shorter time (narrower). The areas are equal because the total change in flux linkage is the same.

QUICK CHECK

Which change would not increase the e.m.f. induced when a magnet is pushed into a coil?

- Using a coil with more turns
- Pushing the magnet in faster
- Using a stronger magnet
- Using a coil made of wire with a higher resistance

The e.m.f. depends only on the rate of change of flux linkage. A higher resistance would reduce the induced **current**, but not the e.m.f.